57. Insert Intervals

The array is composed of 3 subarrays that can all be ∅.

insert((7, 8))

(2, 4), (6, 7), (8, 9), (12, 18)
B M M A

(12, 18)	(8, 9)	(2, 4)
A	M	B

BEFORE curr = (6, 7) 7 < 8

new = (8, 12) curr.end < new.start

MERGE curr = (6, 8) 8 >= 7

new = (7, 12) curr.end >= new.start

AFTER same condition as before

res.extend(intervals[merge_idx + 1:]

Edge Cases: Account for subarrays not missing

before_idx = -1

= i if intervals[i] in BEFORE

merge_idx = before_idx

= i if in MERGE

after_idx = merge_idx + 1

1 pass